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AI IN HEALTHCARE
SYMPOSIUM

Improving Pre-operative Assessment Quality with Dual-Agent AI Health Questionnaires

— a prototype and study proposal

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INTRODUCTION

Accurate pre-operative patient history is essential for surgical safety. Standard questionnaires face a fundamental tradeoff — longer forms are more comprehensive but yield lower completion rates and accuracy. Static forms cannot adapt to individual patient profiles, leading to irrelevant questions and missed clinically important information.

AIMS

To obtain higher quality pre-operative patient history by asking targeted, patient-specific questions based on each patient's profile and prior responses — naturally reducing questionnaire length as a consequence of improved relevance.

METHODS

A dual-agent LLM system dynamically tailors questionnaires to each patient. The system iterates until the history is sufficiently complete. It is model-agnostic, with LLM components interchangeable across foundation models.

Personalised patient greeting

Standard baseline questions

Adaptive follow-up: insulin → diabetes

Agent reasoning trace

Validated question bank enforced

Agent Questions

To prepare for your upcoming procedure, we need to gather some important information about your current health status. Please let us know if you are taking any medications, including prescription drugs, over-the-counter medicines, supplements, or herbal remedies. It's also crucial that we know about any allergies you may have, whether to medications, latex, or other substances. Additionally, we'd like to understand if you have any physical limitations that may affect your ability to move around or care for yourself independently. This information will help us ensure your safety and comfort during the procedure and recovery process.

Have you or a close family member ever had any severe reactions to anaesthesia?

◯ Yes ◯ No

Are you currently taking any medications (including puffers, sprays, patches, natural or herbal medicines and over-the-counter medication)?

◯ Yes ◯ No

Do you have difficulty climbing a flight of stairs (10 or more steps) without the use of aids?

◯ Yes ◯ No

Do you have difficulty walking a block (300 metres or more) by yourself, without the use of aids?

◯ Yes ◯ No

Can you take care of yourself, that is: eat, dress, bathe and use the toilet?

◯ Yes ◯ No

Do you have any allergies to medicines, tapes, latex/rubber, solutions/antiseptics or food?

◯ Yes ◯ No

Next

Chat interface (descending order)

{ "questions": { "medications": "Are you currently taking any medications (including puffers, sprays, patches, natural or herbal medicines and over-the-counter medication)?", "allergies": "Do you have any allergies to medicines, tapes, latex/rubber, solutions/antiseptics or food?", "hasAnaestheticMedicineReactions": "Have you or a close family member ever had any severe reactions to anaesthesia?", "difficultyStairs": "Do you have difficulty climbing a flight of stairs (10 or more steps) without the use of aids?", "difficultyWalkBlock": "Do you have difficulty walking a block (300 metres or more) by yourself, without the use of aids?", "takeCareOfSelf": "Can you take care of yourself, that is: eat, dress, bathe and use the toilet?" }, "rationale": "The selected questions cover the key areas requested: medications the patient is currently taking, any allergies or previous reactions to anesthesia, and functional limitations related to mobility and self-care. These are important factors to consider for anesthesia planning and patient safety.", "patientSummary": "To prepare for your upcoming procedure, we need to gather some important information about your current health status. Please let us know if you are taking any medications, including prescription drugs, over-the-counter medicines, supplements, or herbal remedies. It's also crucial that we know about any allergies you may have, whether to medications, latex, or other substances. Additionally, we'd like to understand if you have any physical limitations that may affect your ability to move around or care for yourself independently. This information will help us ensure your safety and comfort during the procedure and recovery process." }

STEP 1. COLLECT LIST OF MEDICATIONS, ALLERGIES, AND FUNCTIONAL LIMITATIONS. Could you please provide a list of medications that the patient is currently taking? If the patient is not taking any medications, please state "No medications". Also, does the patient have any allergies or previous reactions to anesthesia? If none, please state "No allergies" and "No reactions to anesthesia". Finally, could you describe any physical limitations the patient may have, such as difficulty climbing stairs or being unable to care for themselves? If there are no limitations, please state "No functional limitations".

{ "hqAnswers": [], "previousQuestionKeys": [], "patientTitle": "Mr", "patientLastName": "Screenshots", "patientGender": "Male", "scheduledProcedure": "Oesophagectomy", "procedureCategory": "upperGastro", "surgeryLocation": "Kilmore District Hospital", "anaesthetistId": null, "dateOfBirth": "2009-03-03T00:00:00.000Z", "formattedSurgeryDate": null }

Agent Questions

Based on the medication you are taking, it appears you have diabetes. To better understand your condition and how to manage it during your procedure, we need some additional details. Please let us know if you have been diagnosed with diabetes, and if so, how your diabetes is currently being controlled - through diet, oral medications, insulin injections, or a combination of these methods. Having this information will allow us to properly account for your diabetes and ensure your safety throughout the process.

How is your diabetes controlled - by diet, oral medications, or insulin? (Not available to answer)

Do you have diabetes?

◯ Yes ◯ No

Next

Chat interface (descending order)

{ "questions": { "hasDiabetes": "Do you have diabetes?", "diabetesControl": "How is your diabetes controlled - by diet, oral medications, or insulin?" }, "rationale": "The patient is taking Humulin R Regular insulin, which indicates they have diabetes. The selected questions directly ask about the presence of diabetes and how it is being managed (diet, oral medications, or insulin). This information is important for anesthesia planning and management.", "patientSummary": "Based on the medication you are taking, it appears you have diabetes. To better understand your condition and how to manage it during your procedure, we need some additional details. Please let us know if you have been diagnosed with diabetes, and if so, how your diabetes is currently being controlled - through diet, oral medications, insulin injections, or a combination of these methods. Having this information will allow us to properly account for your diabetes and ensure your safety throughout the process." }

Thank you for providing that information. Based on the answers: Medications: The patient is taking Humulin R Regular insulin daily/multiple times per day, with 1 unit in the morning. Allergies: No allergies reported. Reactions to anesthesia: No previous reactions reported. Functional limitations: - No difficulty climbing stairs or walking longer distances without aids - Unable to take care of themselves for basic activities like eating, dressing, bathing, and using the toilet without assistance STEP 2. IDENTIFY UNDERLYING ISSUES AND RELEVANT QUESTIONS. The medication Humulin R Regular indicates the patient has diabetes. I will ask: 1. Do you have diabetes? 2. How is your diabetes controlled - by diet, oral medications, or insulin? No other underlying issues are clearly indicated from the provided information so far.

{ "hqAnswers": [{ "question": "Have you or a close family member ever had any severe reactions to anaesthesia?", "answer": "No" }, { "question": "Are you currently taking any medications (including puffers, sprays, patches, natural or herbal medicines and over-the-counter medication)?", "answer": "[[\"Medication\",\"Dosage\",\"Frequency\",\"Quantity Per Time\"],[\"Humulin R Regular\",\"\",\"Daily / Multiple times per day\",\"Morning: 1\"]]" }, { "question": "Do you have difficulty climbing a flight of stairs (10 or more steps) without the use of aids?", "answer": "No" }, { "question": "Can you do this with assistance or with aids?", "answer": "Not provided" }, { "question": "Do you have difficulty walking a block (300 metres or more) by yourself, without the use of aids?", "answer": "No" }, { "question": "Can you do this with assistance or with aids?", "answer": "Not provided" }, { "question": "Can you take care of yourself, that is: eat, dress, bathe and use the toilet?", "answer": "No" }, { "question": "Do you have any allergies to medicines, tapes, latex/rubber, solutions/antiseptics or food?", "answer": "No" }], "previousQuestionKeys": ["hasAnaestheticMedicineReactions", "medications", "difficultyStairs", "difficultyStairsWithAids", "difficultyWalkBlock", "difficultyWalkBlockAids", "takeCareOfSelf", "allergies"] }

{ "questions": { "medications": "Are you currently taking any medications (including puffers, sprays, patches, natural or herbal medicines and over-the-counter medication)?", "allergies": "Do you have any allergies to medicines, tapes, latex/rubber, solutions/antiseptics or food?", "hasAnaestheticMedicineReactions": "Have you or a close family member ever had any severe reactions to anaesthesia?", "difficultyStairs": "Do you have difficulty climbing a flight of stairs (10 or more steps) without the use of aids?", "difficultyWalkBlock": "Do you have difficulty walking a block (300 metres or more) by yourself, without the use of aids?", "takeCareOfSelf": "Can you take care of yourself, that is: eat, dress, bathe and use the toilet?" }, "rationale": "The selected questions cover the key areas requested: medications the patient is currently taking, any allergies

RESULTS

A prototype was developed and tested on pre-operative scenarios. In one example, the patient's insulin use automatically triggered targeted follow-up questions about diabetes management, functional capacity, and hypoglycaemic episodes — questions a static form would not have asked. An experienced clinician reviewed the agent dialogues and validated the clinical relevance and appropriateness of the questions asked.

CONCLUSIONS

This proof-of-concept demonstrates that a dual-agent GenAI system can dynamically tailor pre-operative health questionnaires, asking clinically relevant targeted questions while suppressing irrelevant ones. This suggests potential to improve both history completeness and patient experience.

FUTURE WORK

- Prospective clinical trial to measure history quality vs. standard questionnaires
- Multi-language support evaluation
- Integration with hospital pre-admission and EMR systems

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